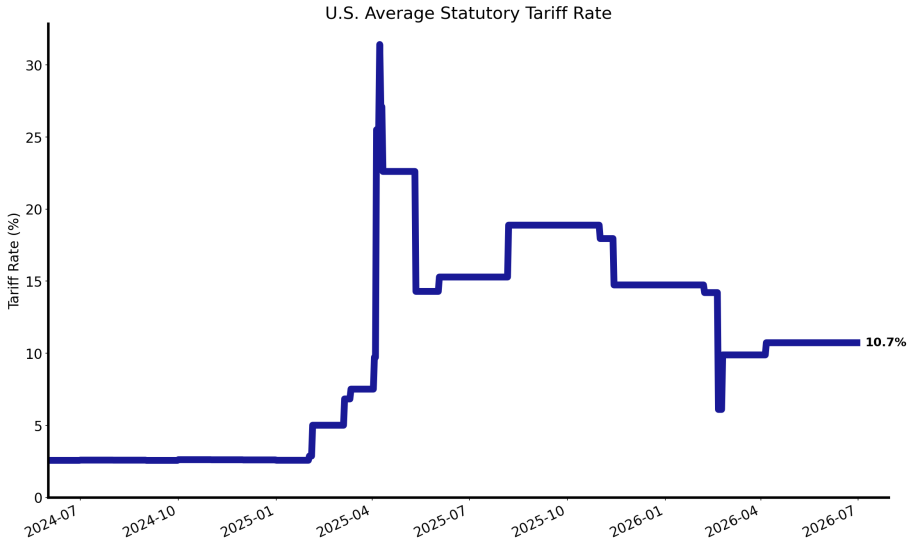
Trade in AI-Related Products

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The views expressed herein are those of the author and not necessarily those of the Federal Reserve Bank of Minneapolis or the Federal Reserve System.

U.S. Tariffs Over the Past Year



Why Everyone Got Trump's Tariffs Wrong

Predictions of the duties' impact on the economy were buoyant or dire. Here's the reality.

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By Chao Deng Follow and Drew An-Pham Follow

Dec. 14, 2025 9:00 pm ET

In the days following "Liberation Day," the contrast between Trump's optimism and more dire predictions from trade experts and economists was stark.

As businesses and consumers tried to make sense of the mixed messages, the president doubled down on promises he'd made during his 2024 presidential campaign. "The markets are going to boom, the stock [market] is going to boom, the country is going to boom," he said on April 3.

Economists and business leaders dialed up predictions of a fallout. BlackRock's Larry Fink said "most CEOs I talk to would say we are probably in a recession right now." JPMorgan Chase said a global recession was even likely.

An economic collapse hasn't materialized. Neither has an economic revival.

The AI Infrastructure Boom and Trade

The U.S. is experiencing a massive AI-investment-driven boom — trillions of dollars invested in data centers and related AI infrastructure.

Question: How is the AI boom influencing international trade flows?

Trade in AI-related products is *the* dominant force in U.S. trade flows over the past two years. Not changes in U.S. trade policy.

Six questions. . .

- How to classify AI-related products?
- What are AI-related products?
- How much trade in AI-related products?
- Where are these products coming from?
- How has trade policy treated AI-related products?
- Does this matter for the aggregate?

How to Classify AI-Related Products?

Use an LLM to classify each of the ∼18,000 HS10 commodity codes by relevance to AI data center infrastructure. Snippet of the prompt:

```
You are an expert in AI data center construction, operations, and supply
chains.  Your task is to classify products by their relevance to building and
operating AI-focused data centers (hyperscale facilities running GPU clusters
for training and inference).

Consider these categories of relevant products:
- COMPUTE: GPUs, CPUs, memory, PCBs, servers, storage drives
- NETWORKING: Fiber optics, switches, routers, transceivers
- COOLING: Chillers, cooling towers, CRAH units, fans, pumps
- ELECTRICAL: Transformers, switchgear, UPS, batteries, generators
- BUILDING: Structural steel, concrete, rebar, raised floors
- SPECIALTY MATERIALS: Rare earths, copper, aluminum...
```

Each HS10 code gets: **Relevance**, **Category**, **Confidence**, and **Reasoning**.

How to Classify AI-Related Products? — An Example

The LLM sees the HS10 description *and* the NAICS industry code:

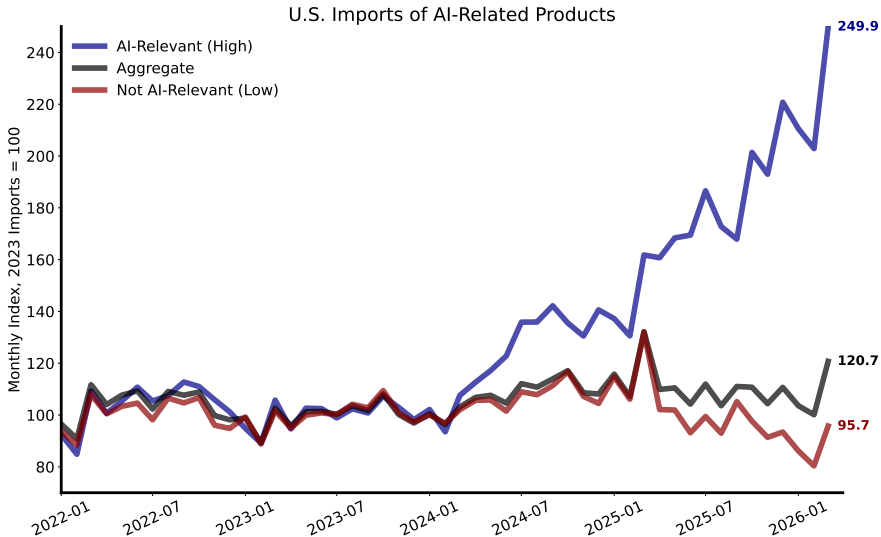
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HS10:  8542310040
Description:  PROCESSORS (INCLUDING MICROPROCESSORS): GRAPHICS PROCESSING UNITS
(GPUS)
NAICS: 334413 --- SEMICONDUCTOR AND RELATED DEVICE MANUFACTURING

RELEVANCE: High   CONFIDENCE: 100%
CATEGORY: Compute Hardware
REASONING: GPUs are the core compute hardware in AI data centers, specifically
designed for parallel processing required in ML training and inference.
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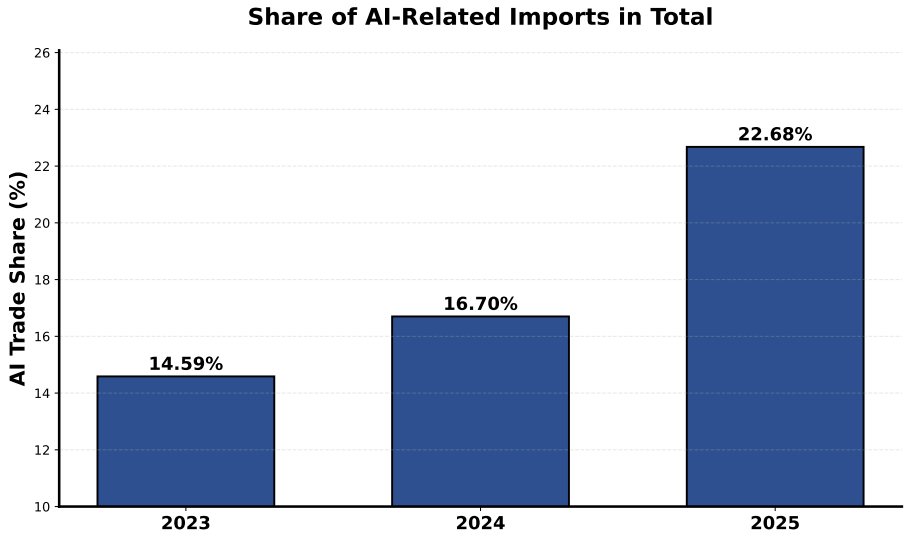
What Are AI-Related Products?

Category	# HS10 Codes	2023 ($B)	2025 ($B)	Change (%)
High AI Relevance	**645**	**379.0**	**654.0**	**+72.6**
Compute Hardware	163	144.4	353.8	+144.9
Electrical Power	250	116.9	141.8	+21.3
Networking Telecom	24	62.9	99.5	+58.2
Cooling HVAC	137	41.5	47.5	+14.4
Building Structure	44	12.1	10.1	-16.5
Fire Safety Security	8	0.6	0.7	+10.5
Specialty Materials	19	0.4	0.5	+22.5
Low AI Relevance	15,915	1,834.7	1,880.9	+2.5
Total Trade	**18,364**	**2,598.3**	**2,883.9**	**+11.0**

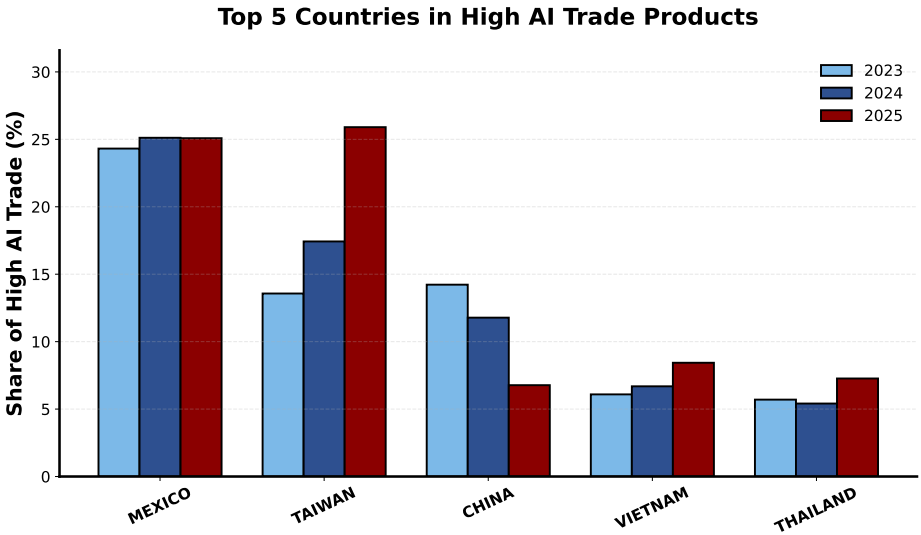
How Much Trade in AI-Related Products? A lot.



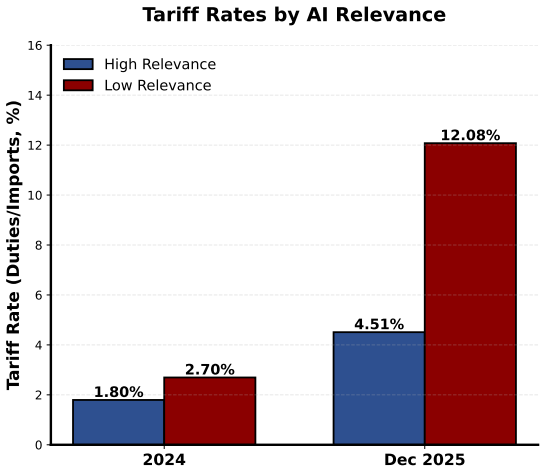
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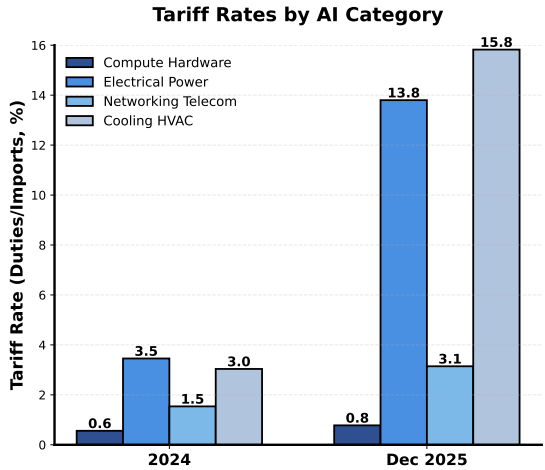


Where Are These Products Coming From?



How Has Trade Policy Treated AI-Related Products?



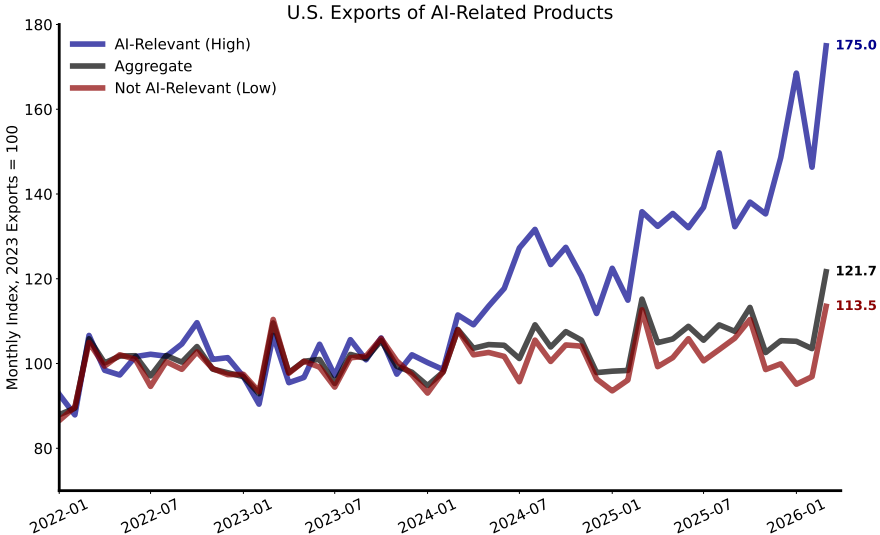


How Has Trade Policy Treated AI-Related Products?

Exemption List	# HS10 Codes	AI High Trade ($B)	Share (%)
Consumer Electronics (EO 14257)	109	424.5	64.9
Annex II (EO 14257)	720	19.4	3.0
Section 122 Surcharge (Feb 2026)	1,098	445.9	68.2
Any Exemption List	—	**448.6**	**68.6**

Nearly **69%** of AI-related imports by value fall on at least one exemption list. Compute Hardware is almost entirely shielded — 97% of its $354B in imports is exempt.

U.S. Exports of AI-Related Products



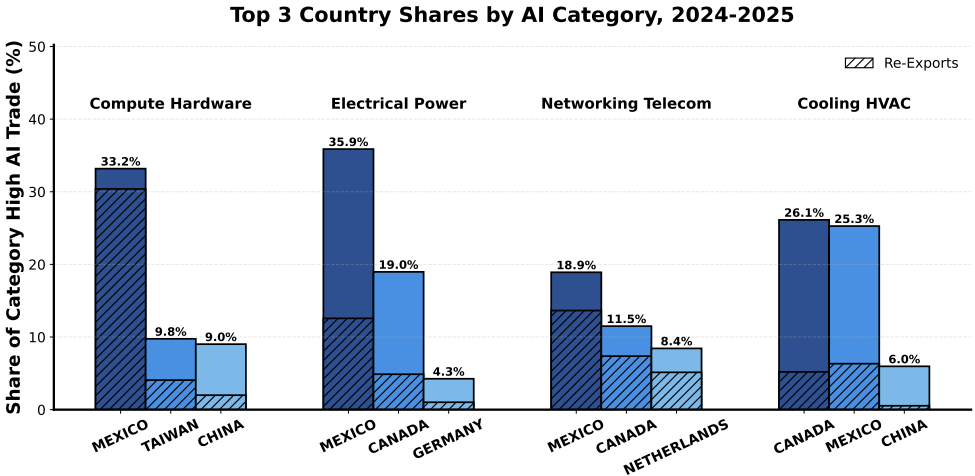
Mexico + Add to myFT

How Mexico became a surprise cornerstone of America's AI boom

Factories churning out servers for data centres are pushing exports to record levels



U.S. Exports of AI-Related Products — Destinations



Does This Matter for the Aggregate? Yes.

A simple accounting exercise:

What would the U.S. goods trade deficit have been if AI-related products had grown the same as non-AI products?

Year	Imports	Exports	Excess AI Imp.	Excess AI Exp.	Net Effect
2023	2,598	1,552	—	—	—
2024	2,792	1,601	66	33	+32
2025	2,884	1,648	265	71	**+194**

Summing Up...

- AI-related products account for **23%** of U.S. imports in 2025.
- Mexico and Taiwan each account for ∼25% of AI trade — Mexico's role is the surprise.
- Effective tariffs on AI products are lower than on other goods — exemption lists cover 69% of AI imports by value.
- The U.S. goods trade deficit would have been $194B smaller absent the AI boom.

Paper, data, and code: https://github.com/mwaugh0328/ai-trade-index

Questions?